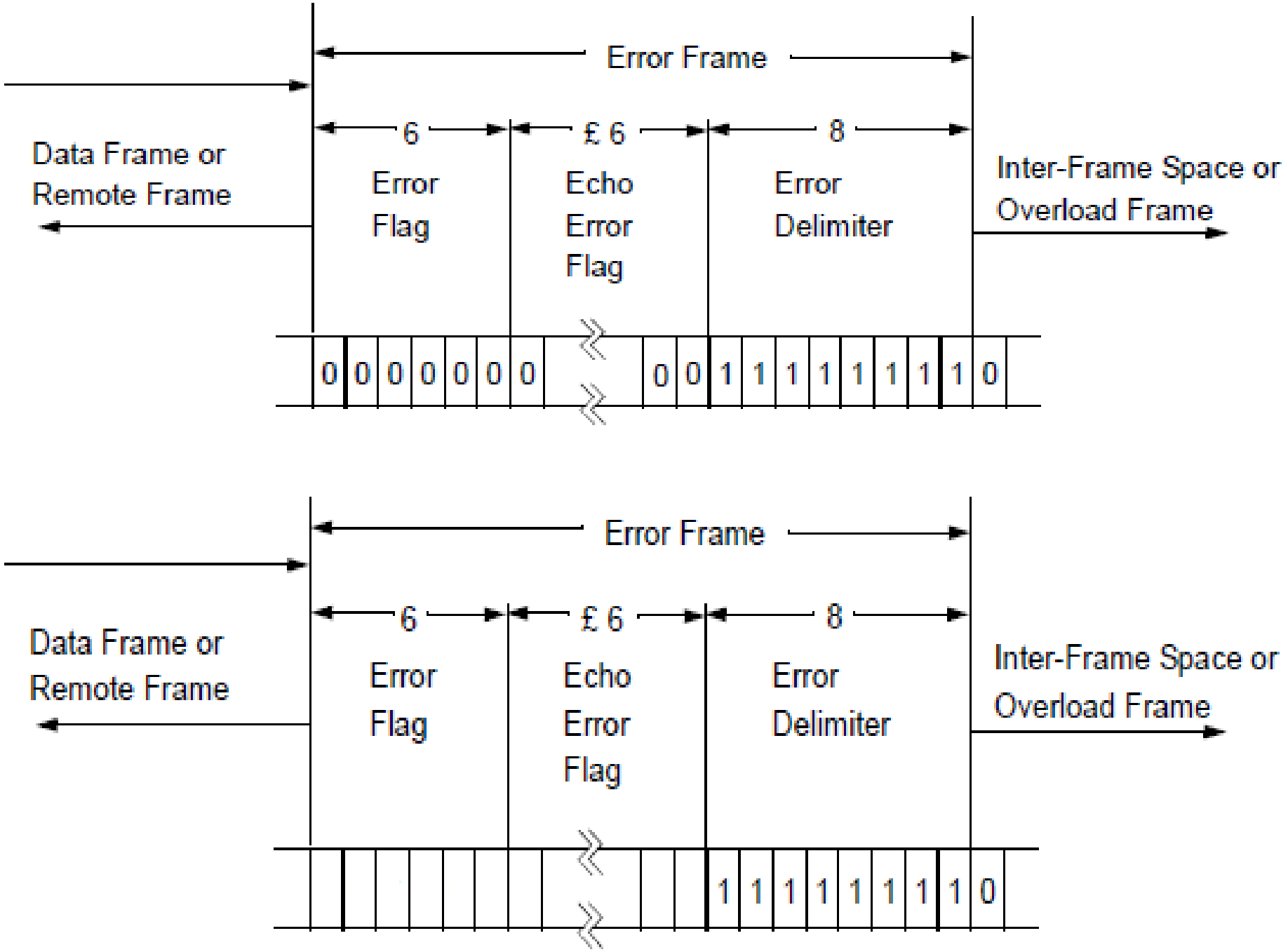
CAN PROTOCOL


Error Frame
6
£ 6
8
Data Frame or
Remote Frame
Error
Flag
Echo
Error
Flag
Error
Delimiter
Inter-Frame Space or
Overload Frame
0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0
Error Frame
6
£ 6
8
Data Frame or
Remote Frame
Error
Flag
Echo
Error
Flag
Error
Delimiter
Inter-Frame Space or
Overload Frame
1 1 1 1 1 1 1 1 0

ERROR DETECTION

- CRC ERROR
- ACKNOWLEGEMENT ERROR
- FORM ERROR
- BIT ERROR
- STUFF ERROR

Error States

Detected errors are made known to all other nodes via error frames. The transmission of the erroneous message is aborted and the frame is repeated as soon as possible. Furthermore, each CAN node is in one of the three error states according to the value of the internal error counters:

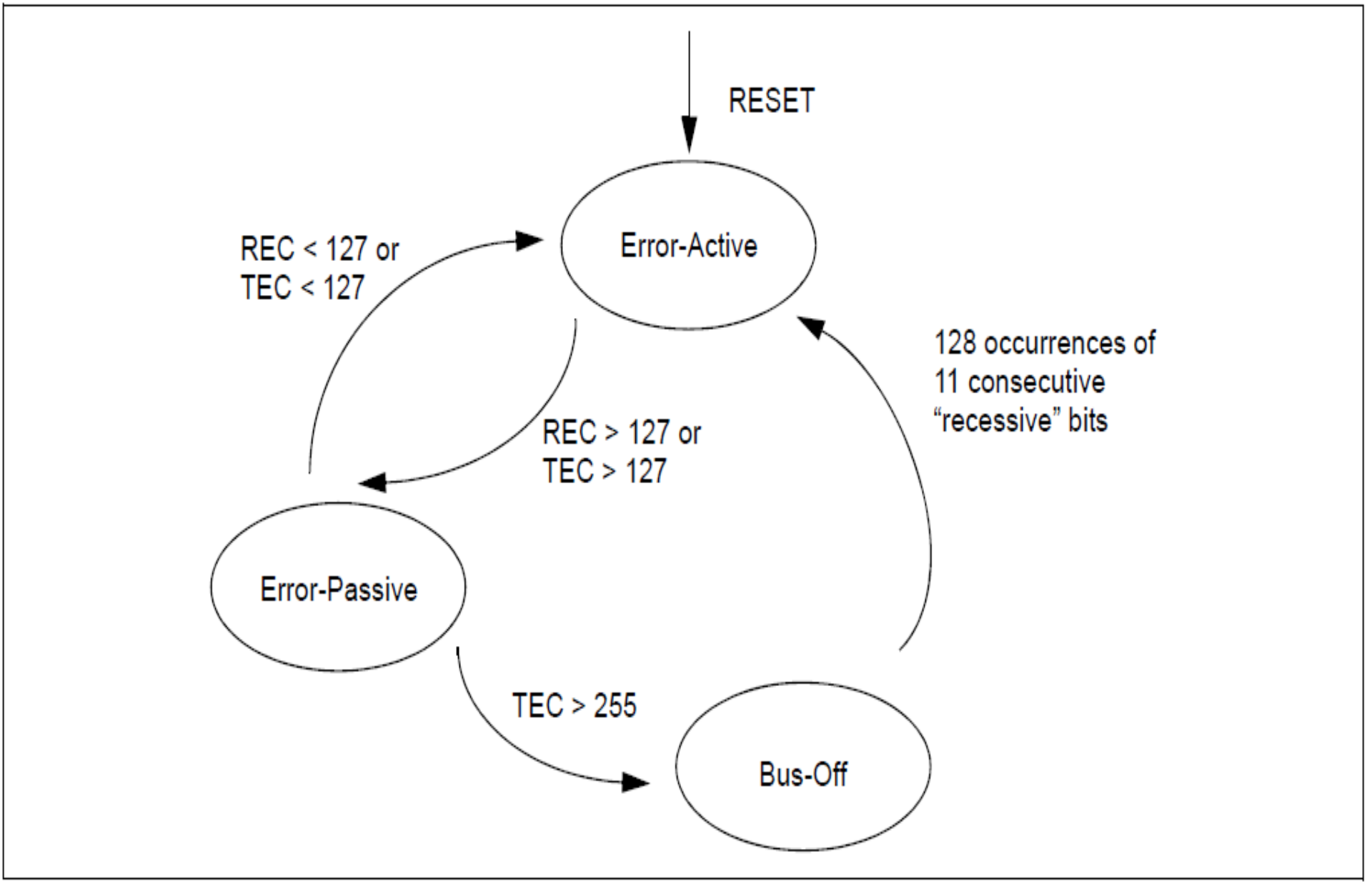
1. Error-active.
2. Error-passive.
3. Bus-off (transmitter only).

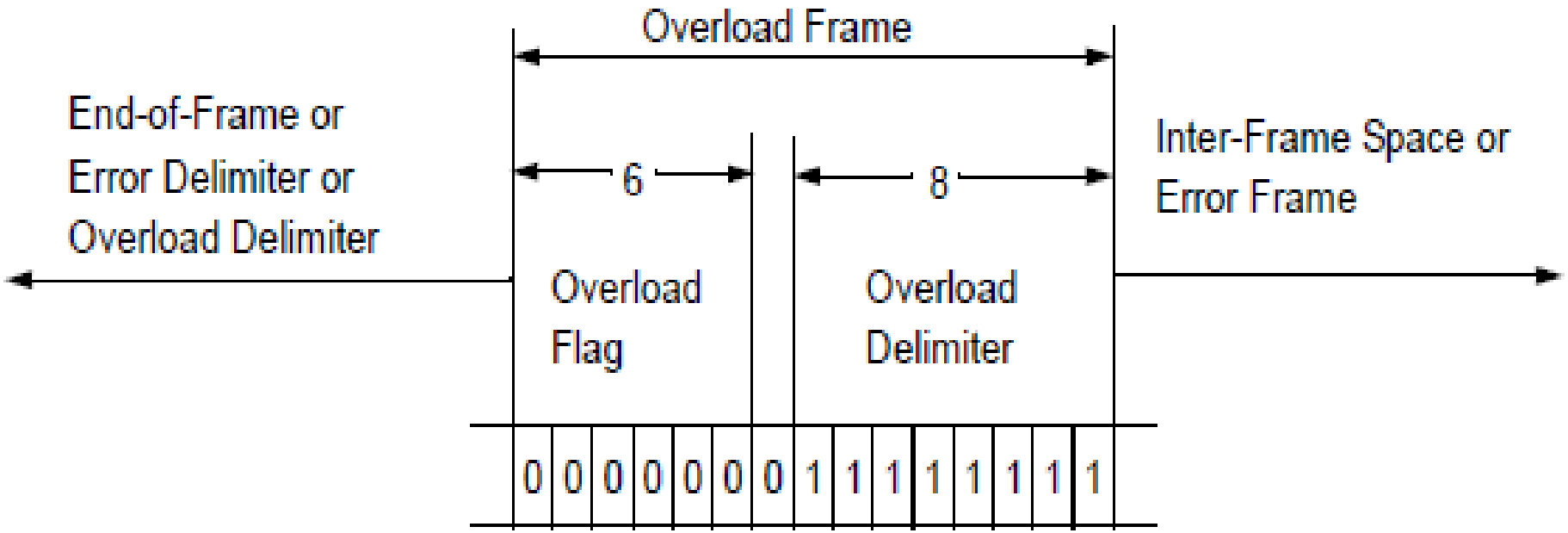
The error-active state is the usual state where the node can transmit messages and active error frames (made of dominant bits) without any restrictions.

In the error-passive state, messages and passive error frames (made of recessive bits) may be transmitted.

The bus-off state makes it temporarily impossible for the station to participate in the bus communication. During this state, messages can neither be received or transmitted. Only transmitters can go bus-off.

ERROR MODES STATE DIAGRAM




Overload Frame
End-of-Frame or
Error Delimiter or
Overload Delimiter
6
8
Inter-Frame Space or
Error Frame
Overload
Flag
Overload
Delimiter
0 0 0 0 0 0 0 1 1 1 1 1 1 1 1